

Generic Degree Two in $JC(2)$: Secant Projectors and Galois Rigidity

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Abstract

We study planar polynomial maps of generic degree two through the scheme of pairs with the same image. For

$$F = (P, Q) : \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2,$$

put $A = \mathbb{C}[x_1, x_2]$, let $I_R(F)$ be the collision ideal in the pair ring $S = \mathbb{C}[x_1, x_2, y_1, y_2]$, and let I_{Δ} be the diagonal ideal. Set $C_F = S/I_R(F)$. Their quotient

$$\text{Obs}(F) = I_{\Delta}/I_R(F) \subseteq C_F$$

is the ideal of the diagonal in the collision ring. When F is Keller, an ordered divided-difference matrix produces an explicit idempotent $p_F \in C_F$ with $\text{Obs}(F) = C_F p_F$.

Writing $B = \mathbb{C}[P, Q] \subseteq A$, $K = \text{Frac}(B)$, and $L = \text{Frac}(A)$, suppose that the induced extension L/K has degree two. Then the generic collision algebra decomposes compatibly with the diagonal as

$$K \otimes_B C_F \simeq_K L \times L,$$

where diagonal evaluation is first projection. The second factor forces $\text{Obs}(F) \neq 0$; when F is Keller, the secant identity $\text{Obs}(F) = C_F p_F$ then gives $p_F \neq 0$. On the other hand, a separable quadratic extension is Galois, and classical Keller–Galois rigidity makes a hypothetical generic-degree-two Keller map a polynomial automorphism. Then $\text{Obs}(F) = 0$ and $p_F = 0$, a contradiction. Thus no planar Keller map has generic degree two. The collision and secant arguments are formalized in Lean; the classical rigidity theorem is isolated there as the sole external mathematical input to the degree-two exclusion.

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1 Introduction

The complex two-dimensional Jacobian conjecture $JC(2)$ asks whether every polynomial map

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF \in \mathbb{C}^{\times},$$

has a polynomial inverse. Such an F is called a *Keller map*. The condition on the Jacobian is local, while polynomial invertibility is global. Collision geometry makes the global issue visible by retaining pairs (x, y) for which $F(x) = F(y)$.

The companion paper [vdV26] develops the dimension-independent collision construction. It associates to a polynomial map two ideals

$$I_R(F) = (F_i(x) - F_i(y)), \quad I_{\Delta} = (x_i - y_i)$$

inside the coordinate ring of ordered pairs. The containment $I_R(F) \subseteq I_{\Delta}$ induces the obstruction ideal

$$\text{Obs}(F) = I_{\Delta}/I_R(F)$$

in the collision ring C_F . This ideal defines the diagonal inside the collision scheme. Its vanishing says that the collision scheme consists only of the diagonal; over $\mathbb{C}$, this is equivalent to polynomial automorphy.

The same construction has a generic field-theoretic form. Write

$$A = \mathbb{C}[x_1, x_2], \quad B = \mathbb{C}[P, Q] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

For a Keller map, L/K is finite and separable, and

$$d_F = [L : K]$$

is the generic degree of F . At the generic point, Paper I gives the compatible isomorphism

$$K \otimes_B C_F \simeq_K L \otimes_K L,$$

with base-changed diagonal evaluation equal to tensor multiplication.

Paper I already uses classical Galois rigidity to exclude generic degrees one and two in every dimension. The purpose here is to realize the quadratic exclusion explicitly in the planar collision ring. A recent preprint states the stronger exclusion of every prime generic degree [Mos24]. We do not use that broader argument; the focus here is the explicit collision certificate and its formal verification.

This paper specializes that construction to $d_F = 2$. There are two complementary descriptions of the collision obstruction. The first is geometric and explicit: ordered divided differences produce the planar multivariate Bézoutian determinant and an idempotent p_F generating $\text{Obs}(F)$. The second is generic and field-theoretic: a separable quadratic extension satisfies

$$L \otimes_K L \simeq_L L \times L,$$

and tensor multiplication is projection onto the marked first factor. For a Keller map, the second factor is a genuine residual generic collision, forcing $p_F \neq 0$.

Classical Galois rigidity gives the opposite conclusion. Since every separable quadratic extension is normal, a hypothetical planar Keller map with $d_F = 2$ would be a polynomial automorphism [Cam73, Raz79, Wri81]. Its collision scheme would then be the diagonal, so $p_F = 0$. The contradiction excludes generic degree two.

Section 2 recalls, without reproving, the part of the general construction needed here. Section 3 constructs the planar secant projector. Section 4 proves quadratic collision nonvanishing, and Section 5 combines it with Galois rigidity. Section 6 records the formalization boundary.

This is a generic-degree-two theorem, not a proof of all of $JC(2)$. The remaining possible counterexamples have nonnormal function-field extensions of generic degree at least three. No normalization, boundary, or inertia construction is needed below.

2 Inputs from Paper I

We use the dimension-independent construction of Paper I [vdV26]. To keep the two manuscripts notationally identical, let

$$F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n,$$

and put

$$\begin{aligned} A &= \mathbb{C}[x_1, \dots, x_n], & S &= \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n], \\ I_R(F) &= (F_i(x) - F_i(y))_{i=1}^n, & I_{\Delta} &= (x_i - y_i)_{i=1}^n, \\ C_F &= S/I_R(F), & R_F &= \text{Spec } C_F. \end{aligned}$$

Since $I_R(F) \subseteq I_{\Delta}$, diagonal substitution descends along the quotient map $q_F: S \twoheadrightarrow C_F$ to

$$\bar{\mu}_F: C_F \twoheadrightarrow A, \quad [s] \longmapsto s(x, x).$$

The image of I_{Δ} is the ideal

$$\text{Obs}(F) := q_F(I_{\Delta}) = I_{\Delta}/I_R(F) \subseteq C_F,$$

which defines the diagonal inside R_F .

The following two results are quoted from Paper I; no proof of the general collision construction is repeated here.

Theorem 2.1 (Diagonal kernel and automorphism detection). *For every complex polynomial map as above,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_{\Delta}/I_R(F).$$

Moreover,

$$F \text{ is a polynomial automorphism} \iff I_R(F) = I_{\Delta} \iff \text{Obs}(F) = 0.$$

The implication $\text{Obs}(F) = 0 \Rightarrow F$ is a polynomial automorphism uses Ax–Grothendieck [Ax69]; below we need only the elementary forward implication.

Let

$$B = \mathbb{C}[F_1, \dots, F_n] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\mu_L: L \otimes_K L \longrightarrow L, \quad \ell \otimes \ell' \longmapsto \ell \ell'$$

be tensor multiplication.

Theorem 2.2 (Generic collision bridge). *If L/K is finite, then*

$$K \otimes_B A \simeq_K L$$

and there is a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L$$

which carries the base-changed diagonal evaluation to μ_L . Furthermore, flatness of the localization $B \rightarrow K$ identifies

$$K \otimes_B \text{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F)$$

as K -vector spaces.

These statements combine the fiber-product, generic-base-change, and diagonal-kernel results of Paper I. Their compatibility with μ_L marks which generic factor is the diagonal.

Return now to

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2.$$

For a Keller map, P, Q are algebraically independent and L/K is finite separable. We write

$$d_F := [L : K]$$

for its generic degree. The 2 in $J\mathcal{C}(2)$ denotes ambient dimension; the separate condition $d_F = 2$ concerns the generic fiber. The nonvanishing argument below only uses a finite separable extension of degree two. The Keller condition enters through the nonzero constant Jacobian and, at the end, through Galois rigidity.

3 The explicit planar secant projector

The matrix below is the ordered multivariate Bézoutian matrix associated to F ; compare [McK23, SS75]. We retain the term *secant matrix* because its determinant separates the diagonal collision sheet from its complement.

Let

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^\times.$$

Set

$$S = \mathbb{C}[x_1, x_2, y_1, y_2], \quad \mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

3.1 Ordered divided differences

For $H \in \mathbb{C}[x_1, x_2]$, define

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1},$$

$$D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients lie in S . The ordered telescoping identity is

$$H(x) - H(y) = D_1 H (x_1 - y_1) + D_2 H (x_2 - y_2).$$

Define

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \quad M_F|_{\Delta} = JF.$$

Let

$$\delta_F := \det(M_F) \in S.$$

Proposition 3.1 (Secant annihilation). *The secant determinant satisfies*

$$\delta_F I_{\Delta} \subseteq I_R(F), \quad \delta_F|_{\Delta} = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by the adjugate matrix gives

$$\delta_F \mathbf{d} = \text{adj}(M_F) \mathbf{r}.$$

Both entries on the right belong to $I_R(F)$. Since the entries of $\mathbf{d}$ generate I_{Δ} , the first assertion follows. Restriction to the diagonal gives

$$\delta_F|_{\Delta} = \det(M_F|_{\Delta}) = \det JF = c.$$

□

3.2 The idempotent

Let $\bar{\delta}_F := q_F(\delta_F)$ be the class of δ_F in C_F , and put

$$e_F := c^{-1} \bar{\delta}_F, \quad p_F := 1 - e_F = 1 - c^{-1} \bar{\delta}_F.$$

The element e_F is the diagonal projector, while p_F is the off-diagonal collision projector.

Theorem 3.2 (Planar secant projector). *With the notation above,*

$$e_F^2 = e_F, \quad p_F^2 = p_F, \quad e_F p_F = 0, \quad e_F + p_F = 1.$$

Moreover,

$$\text{Obs}(F) = C_F p_F, \quad \text{Ann}_{C_F}(\text{Obs}(F)) = C_F e_F,$$

and p_F is the unique idempotent $u \in C_F$ satisfying $\text{Obs}(F) = C_F u$. Hence

$$C_F \simeq A \times C_F p_F.$$

In particular,

$$p_F = 0 \iff \text{Obs}(F) = 0.$$

Proof. Proposition 3.1 gives

$$\bar{\delta}_F \text{Obs}(F) = 0, \quad \bar{\delta}_F - c \in \text{Obs}(F).$$

Equivalently,

$$e_F \text{Obs}(F) = 0, \quad e_F - 1 \in \text{Obs}(F).$$

Hence $e_F(e_F - 1) = 0$, so $e_F^2 = e_F$; the corresponding identities for $p_F = 1 - e_F$ follow.

The second relation says $p_F \in \text{Obs}(F)$. If $j \in \text{Obs}(F)$, then

$$j = (e_F + p_F)j = p_F j,$$

so $\text{Obs}(F) = C_F p_F$. An element annihilates $C_F p_F$ exactly when its p_F -part is zero, giving

$$\text{Ann}_{C_F}(\text{Obs}(F)) = C_F e_F.$$

For uniqueness, let $u^2 = u$ and suppose $C_F u = C_F p_F$. Since $u \in C_F p_F$, one has $u p_F = u$; since $p_F \in C_F u$, one has $u p_F = p_F$. Thus $u = p_F$.

Complementary idempotents give

$$C_F \xrightarrow{\sim} C_F e_F \times C_F p_F, \quad a \mapsto (a e_F, a p_F).$$

Diagonal evaluation restricts to an isomorphism $C_F e_F \simeq A$: if $a = \bar{\mu}_F(r)$, then $a = \bar{\mu}_F(r e_F)$, while its kernel is $C_F e_F \cap \text{Obs}(F) = 0$. Thus $C_F \simeq A \times C_F p_F$. Finally, $\text{Obs}(F) = C_F p_F$ gives

$$p_F = 0 \iff \text{Obs}(F) = 0.$$

□

Abstractly, the existence of an idempotent splitting is expected: a Keller map is unramified, so its diagonal in the collision scheme is open, while a morphism between affine schemes has closed diagonal [The26b, The26a]. The theorem supplies the explicit Bézoutian representative and identifies the ideal it generates. The uniqueness just proved shows that the resulting p_F is independent of the ordering used to write the divided differences.

4 Quadratic generic collisions

We now work at the generic point. The essential field-theoretic fact is the degree-two marked-root decomposition.

4.1 The quadratic self-tensor product

Let L/K be finite separable of degree two. Paper I's marked-root tensor decomposition applies to a primitive generator α : it splits the marked root from the residual polynomial

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Here h_α has degree one, so its quotient algebra is another copy of L . Thus the general result of Paper I [vdV26] specializes as follows.

Proposition 4.1 (Quadratic marked-root decomposition). *There is an L -algebra isomorphism*

$$\Psi_\alpha: L \otimes_K L \xrightarrow{\sim} L \times L$$

under which tensor multiplication

$$\mu_L: L \otimes_K L \longrightarrow L$$

is first projection. Consequently,

$$\Psi_\alpha(\ker(\mu_L)) = 0 \times L \neq 0.$$

4.2 Nonvanishing of the affine obstruction

Combining the quoted generic collision bridge with Proposition 4.1 transports the nonzero second factor back to the affine obstruction.

Theorem 4.2 (Quadratic collision nonvanishing). *Let $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ be a polynomial map whose induced extension L/K is finite separable of degree two. Then*

$$\text{Obs}(F) \neq 0.$$

Equivalently,

$$I_R(F) \subsetneq I_\Delta.$$

Proof. Flatness of $B \rightarrow K$ gives a canonical identification

$$K \otimes_B \text{Obs}(F) \simeq \ker(1 \otimes \bar{\mu}_F).$$

Under the compatible quadratic product, the right-hand side is

$$\ker(L \times L \xrightarrow{\text{pr}_1} L) = 0 \times L,$$

which is nonzero. Therefore $K \otimes_B \text{Obs}(F) \neq 0$, and hence $\text{Obs}(F) \neq 0$. Since $\text{Obs}(F) = I_\Delta / I_R(F)$, this is equivalent to strict containment. $\square$

This theorem is independent of the Keller condition: generic degree two already forces a residual generic collision. For a planar Keller map, the secant projector makes that factor explicit.

Corollary 4.3 (Nonzero planar projector in generic degree two). *Let*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^\times.$$

If $d_F = 2$, then

$$p_F \neq 0.$$

Proof. The Keller condition makes L/K finite separable. Theorem 4.2 gives $\text{Obs}(F) \neq 0$, and Theorem 3.2 gives $\text{Obs}(F) = C_F p_F$. Therefore $p_F \neq 0$. $\square$

5 Galois rigidity and the collision contradiction

We now use the classical Galois case of the Jacobian conjecture.

Theorem 5.1 (Classical Keller–Galois rigidity). *Let*

$$F: \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

be a Keller map. If the induced finite separable extension L/K is normal, equivalently Galois, then F is a polynomial automorphism.

This is the classical normal/Galois case [Cam73, Raz79, Wri81]; we use it as a literature theorem and do not reprove it here.

Every separable extension of degree two is normal: if the minimal polynomial of a primitive generator has one root in L , its other root also lies in L . Thus Theorem 5.1 applies automatically on the generic-degree-two Keller locus.

Proposition 5.2 (Rigidity forces collision vanishing). *Let*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF = c \in \mathbb{C}^{\times}.$$

If $d_F = 2$, then Keller–Galois rigidity gives

$$\text{Obs}(F) = 0, \quad p_F = 0.$$

Proof. The extension L/K is finite separable of degree two, hence Galois. Theorem 5.1 makes F a polynomial automorphism. Theorem 2.1 then gives

$$I_R(F) = I_{\Delta}, \quad \text{Obs}(F) = 0.$$

Finally, Theorem 3.2 gives $p_F = 0$. □

Theorem 5.3 (Exclusion of generic degree two). *No planar Keller map has generic degree two. Equivalently, every polynomial map*

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2, \quad \det JF \in \mathbb{C}^{\times},$$

satisfies

$$d_F \neq 2.$$

Proof. Suppose that $d_F = 2$. Corollary 4.3, which reads the residual factor in

$$K \otimes_B C_F \simeq_K L \times L,$$

gives $p_F \neq 0$. Proposition 5.2 gives $p_F = 0$. This is a contradiction. □

Equivalently, the two halves of the proof say

$$d_F = 2 \implies \text{Obs}(F) \neq 0$$

by the quadratic generic collision product, while

$$F \text{ Keller and } L/K \text{ normal} \implies \text{Obs}(F) = 0$$

by Galois rigidity and automorphism detection. The secant construction expresses both statements as nonvanishing and vanishing of one canonical idempotent.

6 Lean formalization status

Lean correspondence. The focused theorem spine is formalized in Lean 4.28.0 and Mathlib v4.28.0 [dMU21, The20]. The general collision objects are defined in `CollisionIdeals/General/Collision.lean`. The function-field localization and generic collision bridge are in `CollisionIdeals/General/GenericFiber/FunctionField.lean` and `CollisionIdeals/General/GenericFiber/CollisionDiagonal.lean`. The degree-two marked-root calculation is proved in `CollisionIdeals/General/GenericFiber/QuadraticDecomposition.lean`. Its composition with the polynomial generic-collision bridge, including

$$K \otimes_B C_F \simeq_K L \times L$$

and the descent to $\text{Obs}(F) \neq 0$, is in `CollisionIdeals/General/GenericFiber/QuadraticCollision.lean`. The coefficientwise divided differences, secant determinant, annihilation relation, diagonal restriction, and canonical projector are proved in `CollisionIdeals/Planar/ExplicitSecant.lean`. The planar nonvanishing and vanishing statements are in `CollisionIdeals/Planar/GenericDegreeTwo/Nonvanishing.lean` and `CollisionIdeals/Planar/GenericDegreeTwo/Vanishing.lean`. Their collision-native contradiction and $d_F \neq 2$ are assembled in `CollisionIdeals/Planar/GenericDegreeTwo/Main.lean`. The two Palomar-facing declarations are `CollisionIdeals.Palomar.PaperTwo.explicitPlanarSecantProjector` and `CollisionIdeals.Palomar.PaperTwo.quadraticPlanarCollisionRigidity`. They are checked by `lake build CollisionIdeals.Palomar.PaperTwo.Solution`. The classical theorem in Theorem 5.1 is neither installed as a Lean axiom nor proved in the formalization. Instead, `CollisionIdeals/General/Automorphism/GaloisRigidity.lean` defines the proposition

$$\text{ComplexKellerGaloisRigidity } 2$$

and passes a proof of it explicitly to every theorem using the literature input. Lean therefore proves the degree-two exclusion relative to this single named hypothesis. No normalization, ramification, boundary, purity, or finite-étale interface occurs in the focused proof. Those constructions belong to the separate problem of nonnormal generic degree at least three.

7 Conclusion

For a planar Keller map, the divided-difference determinant gives

$$C_F \simeq A \times C_F p_F, \quad \text{Obs}(F) = C_F p_F,$$

where p_F is the explicit off-diagonal collision projector. Generic degree two forces

$$K \otimes_B C_F \simeq_K L \times L,$$

with diagonal evaluation equal to first projection, so $p_F \neq 0$.

The quadratic extension is nevertheless Galois. Classical Keller–Galois rigidity would make the map a polynomial automorphism, forcing the collision scheme to equal its diagonal and $p_F = 0$. This incompatibility proves that no planar Keller map has generic degree two.

The remaining possible counterexamples to $J\mathcal{C}(2)$ have nonnormal function-field extensions of generic degree at least three. The degree-two case itself is complete once the classical rigidity theorem is taken as input.

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